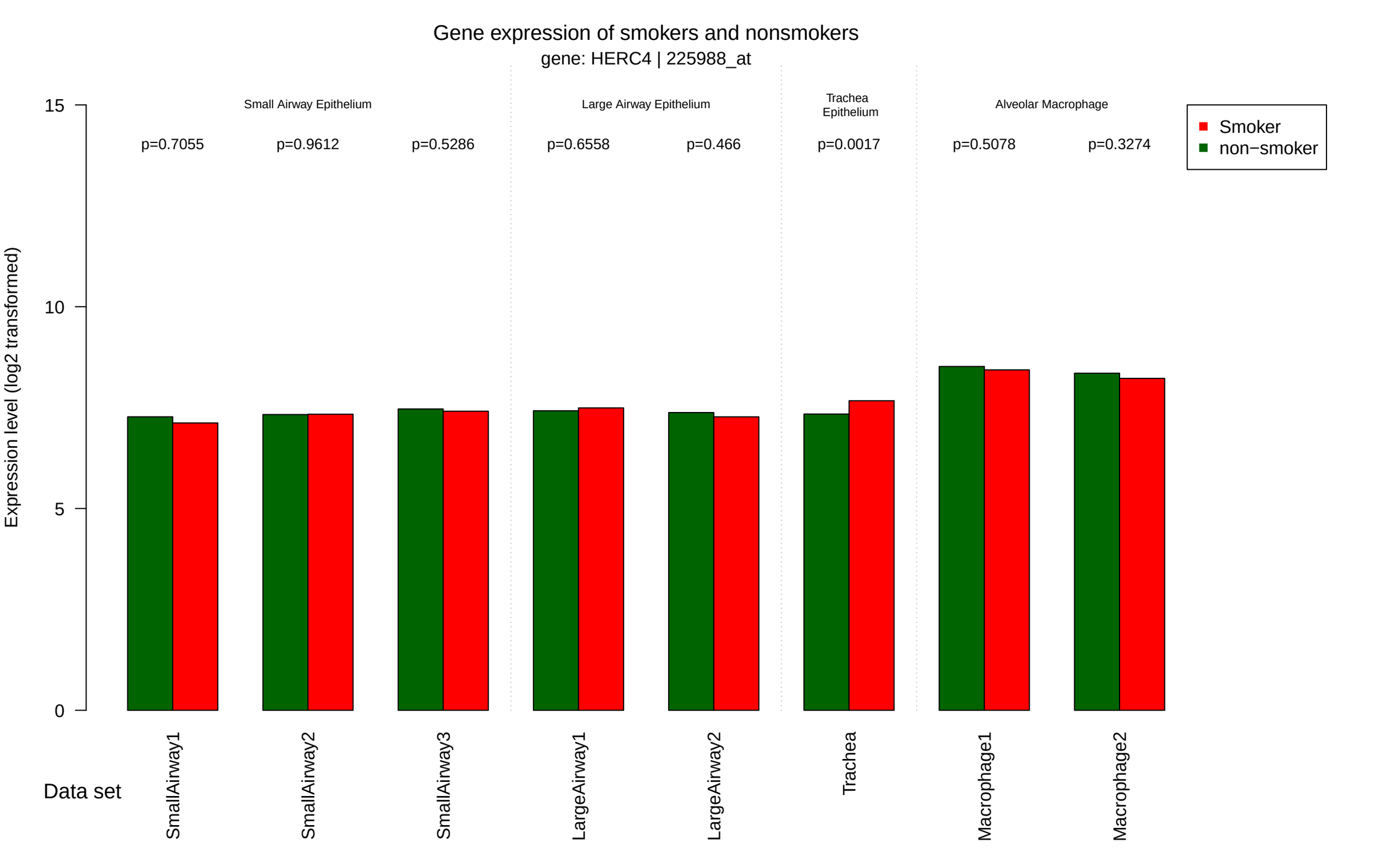
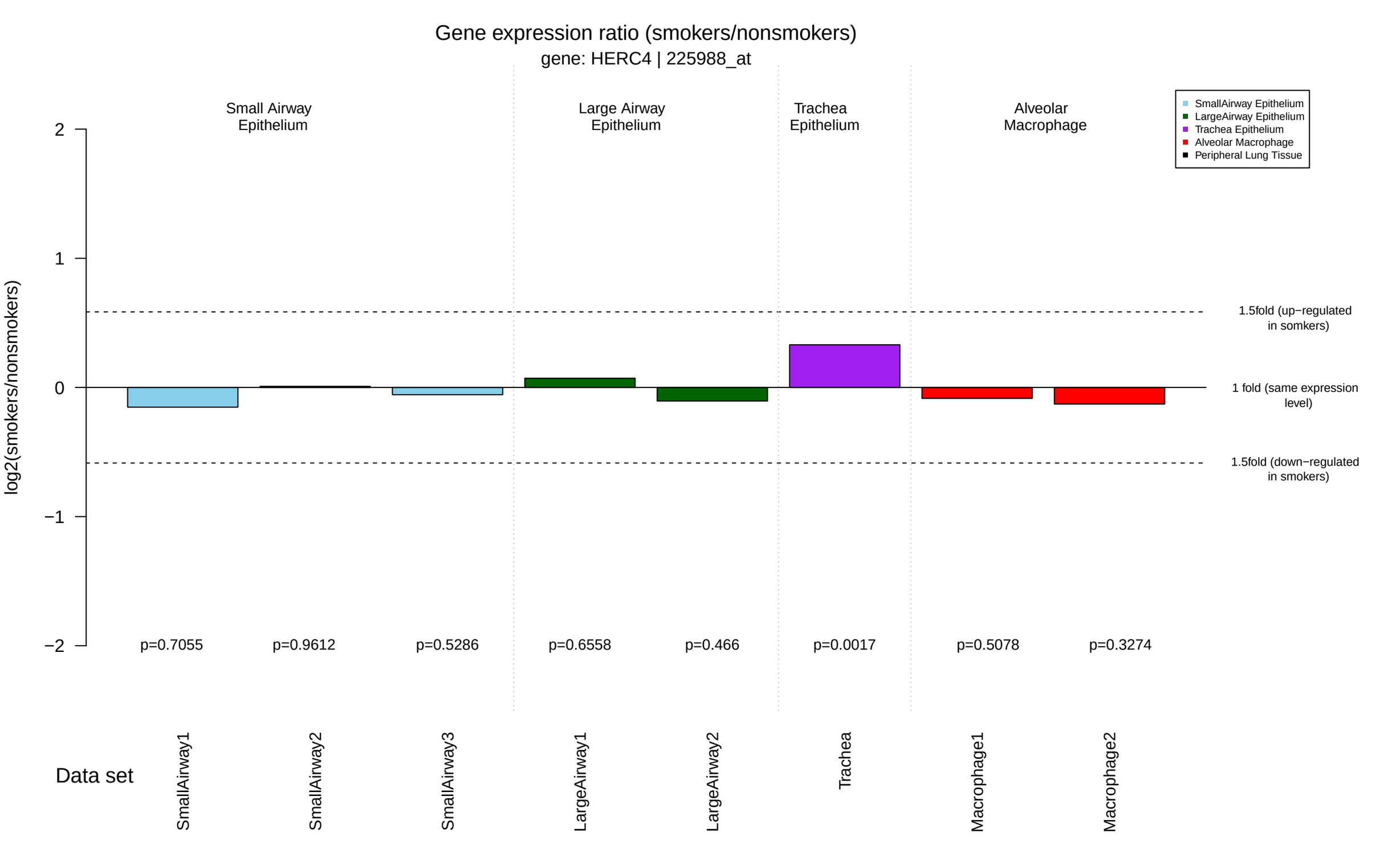
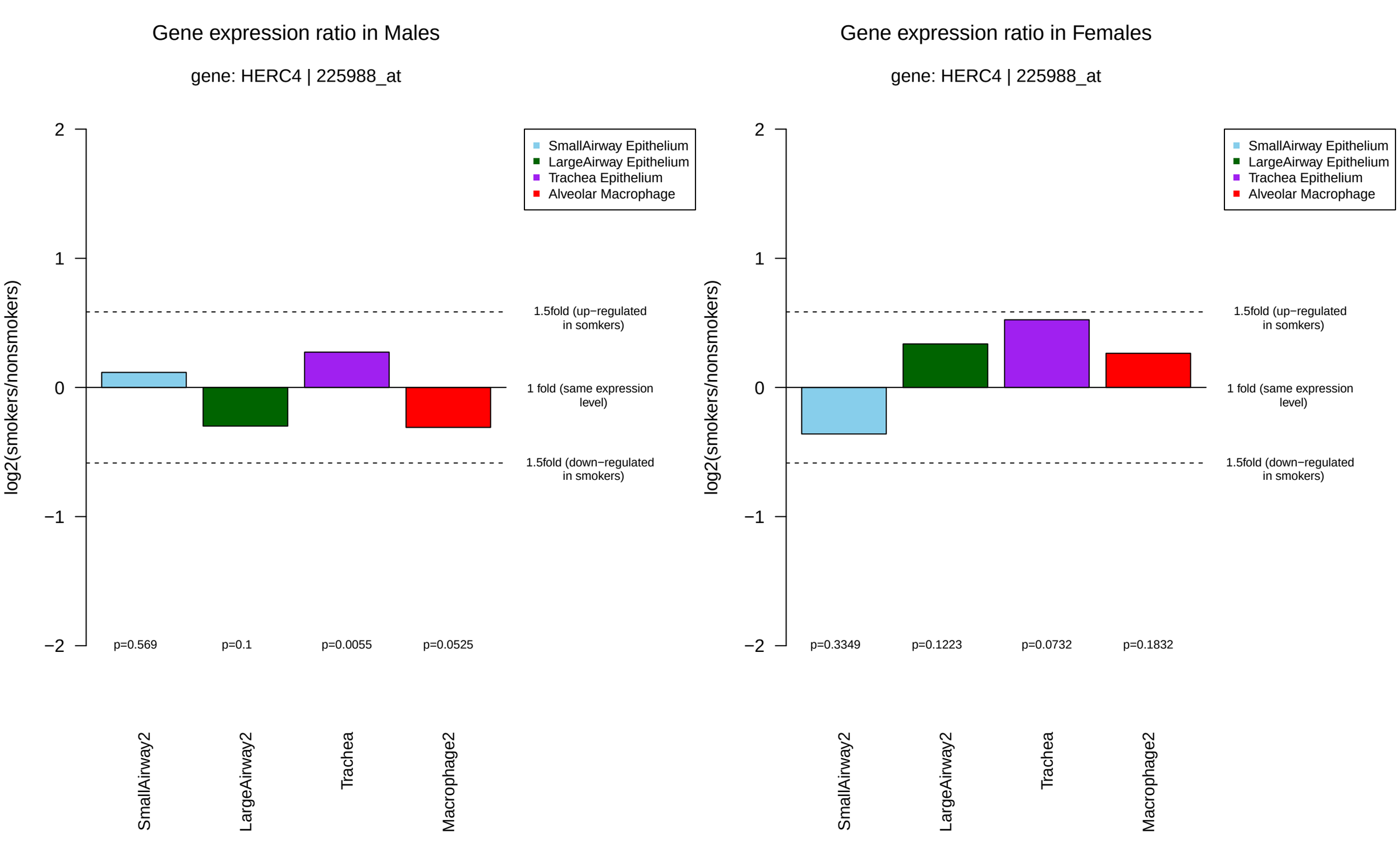
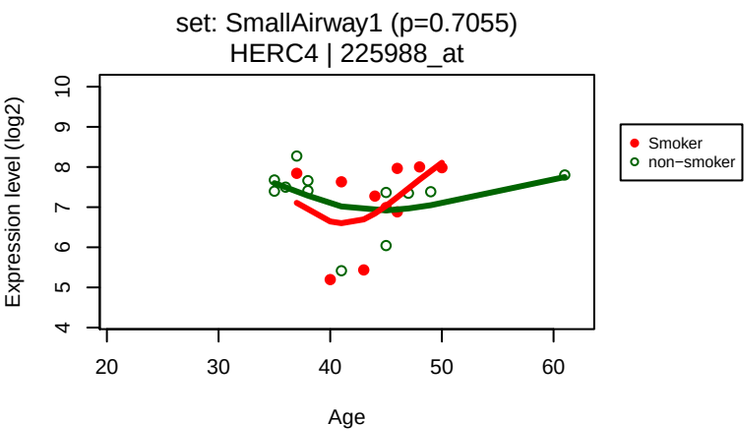
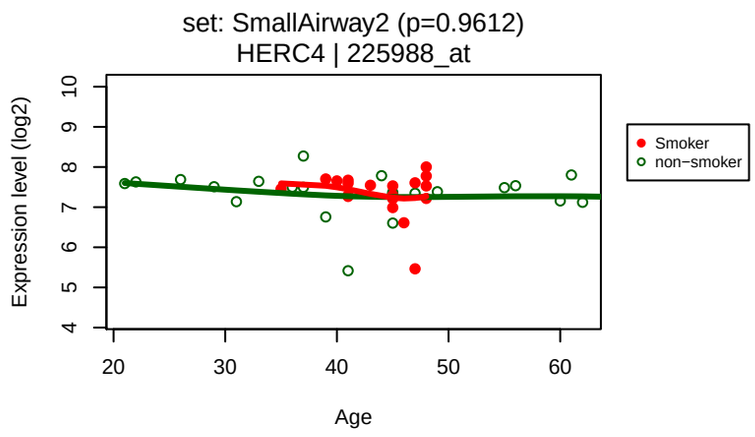
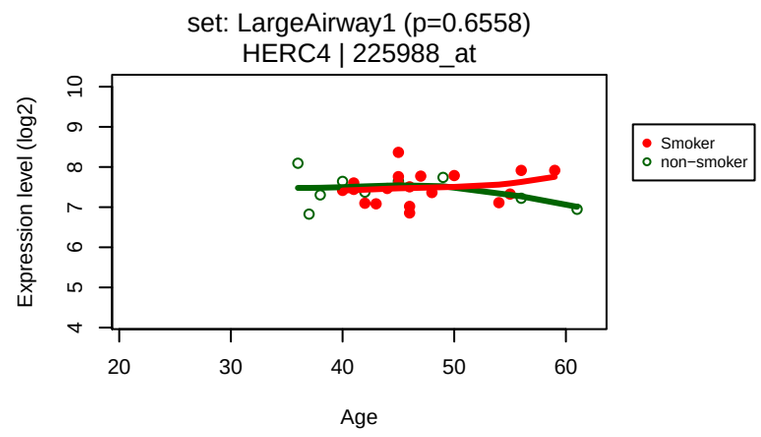

Gene expression of smokers and nonsmokers
gene: HERC4 | 225988_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7055
p=0.9612
p=0.5286
p=0.6558
p=0.466
p=0.0017
p=0.5078
p=0.3274
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

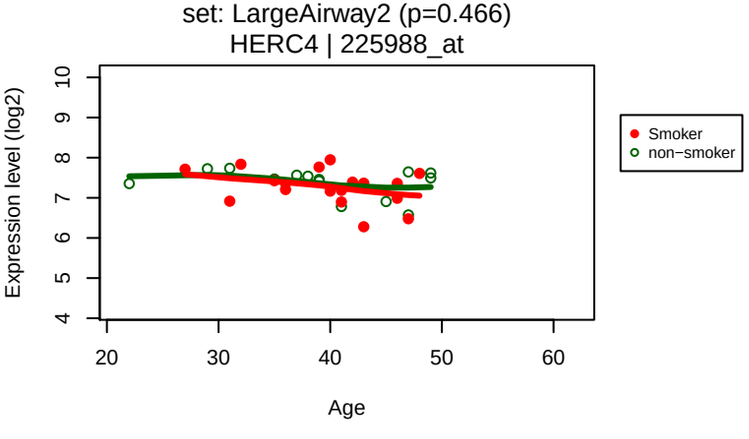

Gene expression ratio (smokers/nonsmokers)
gene: HERC4 | 225988_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7055
p=0.9612
p=0.5286
p=0.6558
p=0.466
p=0.0017
p=0.5078
p=0.3274
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

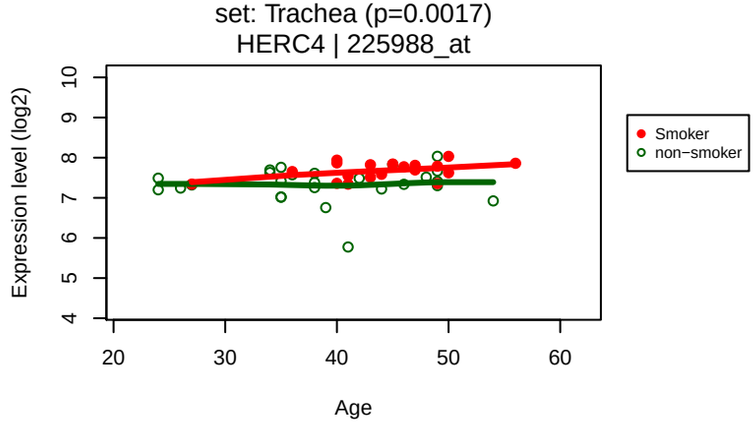

Gene expression ratio in Males
gene: HERC4 | 225988_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.569
p=0.1
p=0.0055
p=0.0525
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HERC4 | 225988_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3349
p=0.1223
p=0.0732
p=0.1832
SmallAirway2
LargeAirway2
Trachea
Macrophage2

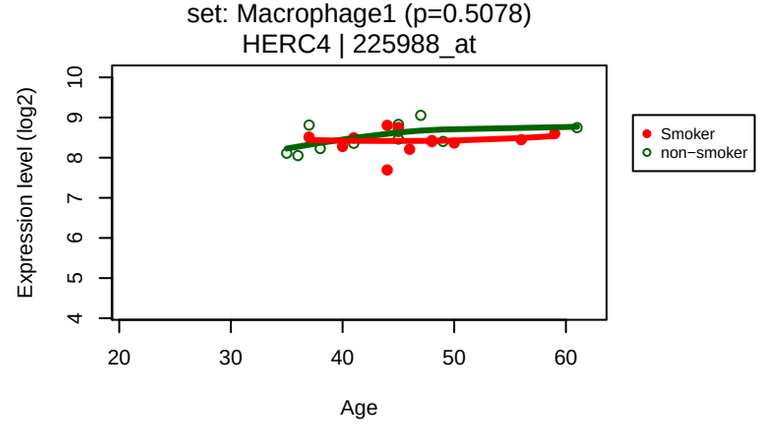

set: SmallAirway1 (p=0.7055)
HERC4 | 225988_at
Expression level (log2)
Age
Smoker
non−smoker

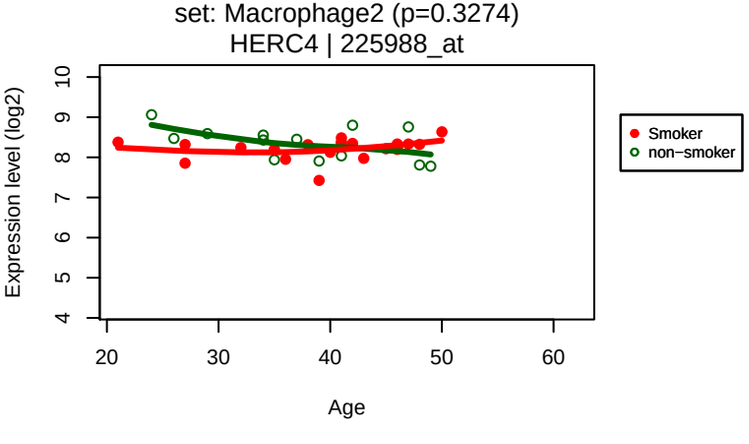

set: SmallAirway2 (p=0.9612)
HERC4 | 225988_at
Expression level (log2)
Age
Smoker
non−smoker

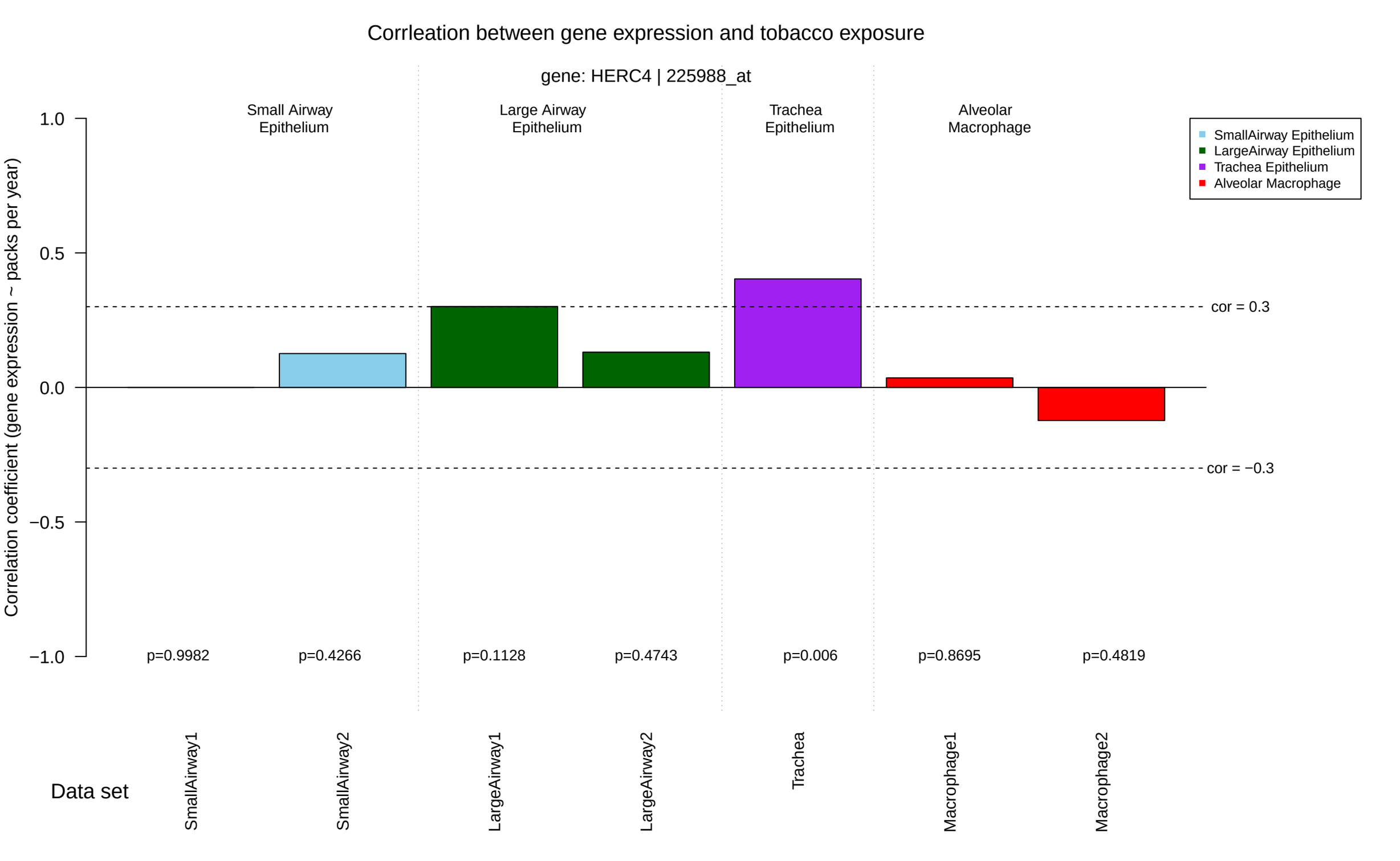

set: LargeAirway1 (p=0.6558)
HERC4 | 225988_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.466)
HERC4 | 225988_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0017)
HERC4 | 225988_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5078)
HERC4 | 225988_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3274)
HERC4 | 225988_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HERC4 | 225988_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9982
p=0.4266
p=0.1128
p=0.4743
p=0.006
p=0.8695
p=0.4819
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2